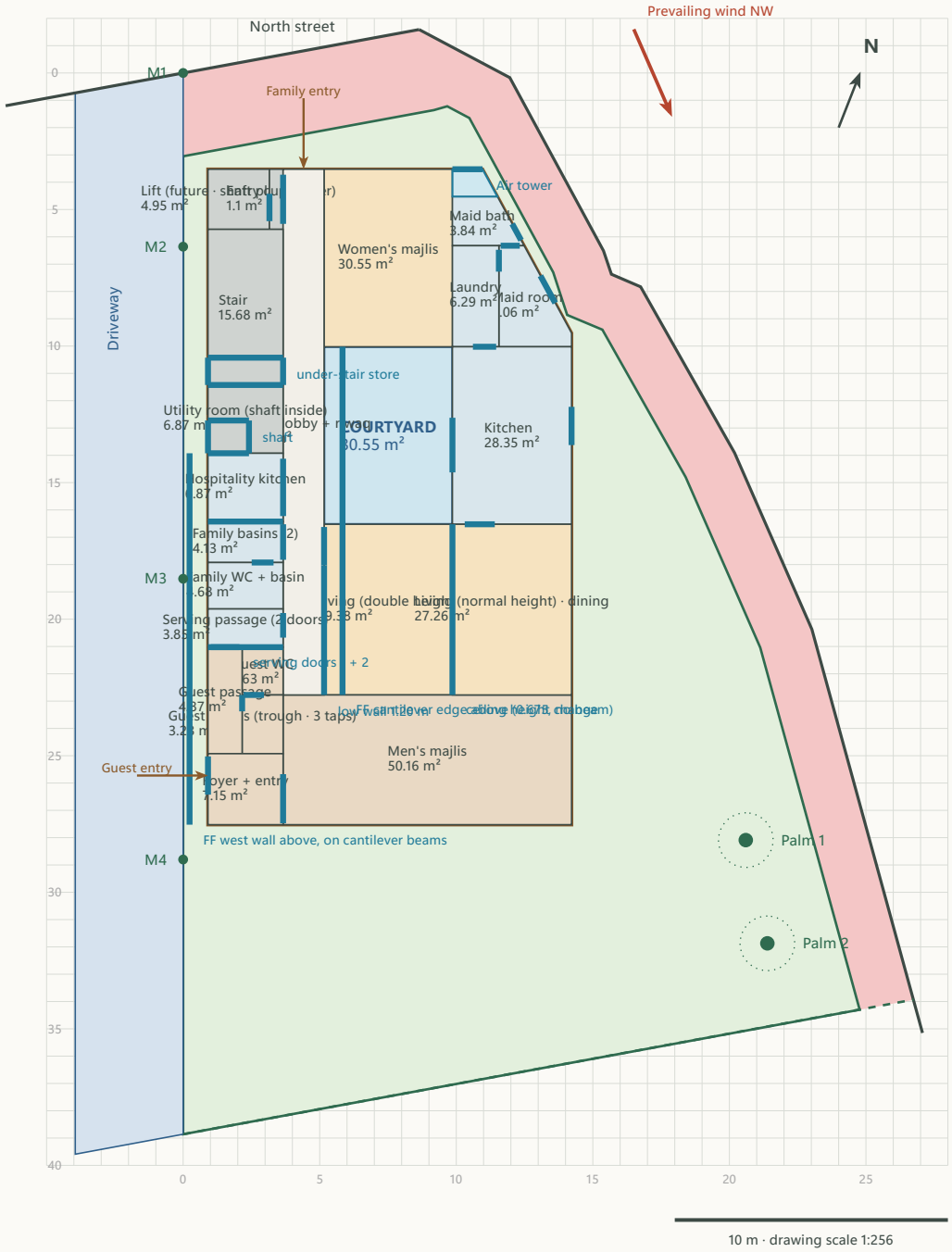


Taif / Palm House

Ground floor V5.1 · pentagon Z4 (V4.0 + first-floor coordination, no columns)

24 Oct 2026 · Ground floor = V4.0 with the east duct removed; V5.1: the three V5.0 projection columns removed · base.typ + design_gf.typ + data/rooms_gf.csv · room studies in studies/ · full description docs/GROUND_FLOOR_V4.md

Ground floor V5.1 (V4.0 + the first-floor coordination: no east duct; **no columns**: the FF projection is split, 0.675 m inward as a plain cantilever and 0.675 m outward over the west wall, the FF west face 0.11 m from the M line) on site-test option Z4: a pentagon 13.311 × 24.0 m on wall centre lines (13.561 × 24.25 outside), the north-east corner cut parallel to the lane down to y 6.0, then the east wall straight south. Outer west wall face 0.79 m from the M line; nearest outer wall face to the envelope 0.05 m (to be confirmed by the official survey). Both palms lie beyond the south facade. Wall allowances 250 mm outer, 200 mm inner, 100 mm dry block partitions. **Middle band** 4.50 m clear: **women's majlis** 4.70 × 6.5 · **courtyard** 4.70 × 6.5 (4.50 × 6.30 clear) · **living (double height)** 4.70 × 6.25, open to the riwaq through a 1.40 m opening, then a 1.20 m low partition. **East service block** (studies/east_wing.typ): **kitchen** 4.361 × 6.5, two parallel counter runs, no island, sink on the courtyard wall, hob and hood on the outside east wall, hood extract straight out through the east wall (V5.0: no east duct); **laundry** 1.70 × 3.7, ≈ 0.95 m clear beside the machines; **maid room** ≈ 6.0 m² clear and **maid bath** 1.8 m deep (≈ 3.0 m² clear), both up to the sloped wall; **air tower** 1.0 m deep (≈ 0.9 m² clear); maid access only kitchen → laundry → maid room → maid bath. **West band:** entry cupboard and **future lift** (2.25 × 2.2, shaft placeholder, a store until then), stair (open, no door), utility room with the service shaft, then the **west wash block** (2.75 × 13.6, studies/guest_wing.typ): **hospitality kitchen** 2.75 × 2.5 (light kitchen, open to the riwaq), a full-height wall, family basins, family WC (no shower), **serving passage** 2.75 × 1.4 (1.20 m clear, two doors never open together, both locked from the family side), then the guest side: guest WC 1.5 × 1.75, **guest basins** (one trough, 3 taps), guest passage, **foyer** with a 1.40 m double door at the north end of its west wall and a 1.80 m opening into the men's majlis at the south end of its east wall. **Family lobby + riwaq** 1.5 × 19.25 · **living (normal height) · dining** 4.361 × 6.25 · **men's majlis** 10.561 × 4.75. Blue outlines inside rooms are the shaft and the under-stair store (not extra area); blue lines mark the ceiling-height change, the FF cantilever edge over the courtyard (x 4.925), the FF west wall above (x −0.675, outside this floor), the low wall and the doors, openings and walls of both blocks.

✓ **All checks passed**

Areas

Envelope	**667.87 m²**
Building limit (envelope unless set)	**309.68 m²**
Other open voids	**0 m²**
Courtyard (open to sky)	**30.55 m²**
Rooms placed (21)	**277.76 m²**
Limit minus courtyard and voids	**279.13 m²**
Air channel + duct	**1.37 m²**
Not yet assigned	**1.37 m²**

Orientation

True compass vs the M line	**21.4°**
NW wind vs the M line	**23.6°**

Concept only. Wind: Global Wind Atlas (area data, 10 m), NW ≈ 15–20 %, SE ≈ 7–8 %. Confirm on site with a ribbon test.

Schedules

All coordinates in metres from M1: x east (square to the M line), y along the M line southwards.

Envelope corners

Corner	x	y
V1	0	38.86
V2	0	3.05
V3	9.18	1.36
V4	9.67	1.22
V5	10.48	1.65
V6	13.55	7.3
V7	14.06	8.86
V8	15.35	9.4
V9	18.4	14.8
V10	21.13	21.04
V11	24.77	34.3

Rooms

Name	Frame	w × d (m)	Area
Women's majlis	M	4.7 × 6.5	30.55 m²
Living (double height)	M	4.7 × 6.25	29.38 m²
Men's majlis	M	10.561 × 4.75	50.16 m²
Maid bath	M	2.622 × 1.8	3.84 m²
Maid room	M	2.661 × 3.7	7.06 m²
Laundry	M	1.7 × 3.7	6.29 m²
Kitchen	M	4.361 × 6.5	28.35 m²
Living (normal height) · dining	M	4.361 × 6.25	27.26 m²
Lift (future · shaft placeholder)	M	2.25 × 2.2	4.95 m²
Entry cupboard	M	0.5 × 2.2	1.1 m²
Family lobby + riwaq	M	1.5 × 19.25	28.87 m²
Stair	M	2.75 × 5.7	15.68 m²
Utility room (shaft inside)	M	2.75 × 2.5	6.87 m²
Hospitality kitchen	M	2.75 × 2.5	6.87 m²
Family basins (2)	M	2.75 × 1.5	4.13 m²
Family WC + basin	M	2.75 × 1.7	4.68 m²
Serving passage (2 doors)	M	2.75 × 1.4	3.85 m²
Guest WC	M	1.5 × 1.75	2.63 m²
Guest basins (trough · 3 taps)	M	1.5 × 2.15	3.23 m²
Guest passage	M	1.25 × 3.9	4.87 m²
Foyer + entry	M	2.75 × 2.6	7.15 m²

Design idea and philosophy

Taif / Palm House · Ground floor V5.1 · pentagon with the north-east corner cut

A house around a cool heart

The house is organised around one idea: a small, shaded courtyard at its centre that stays cool through the day. The rooms the family uses most open onto it. The wide double-height living room, which also holds the dining, faces it from the south; the kitchen from the east; the women's majlis from the north. On the west, the riwaq runs along it with windows. The courtyard is not a leftover space between rooms; it is the room the rest of the house looks into.

Looking in, not out

The outer walls stay quiet, with small openings, as in the traditional houses of the region. Light, air and view come from the inside. This gives privacy from the street and the lane, protects the rooms from the hot sun and dust, and lets the family live openly within their own walls.

Air before sun

Taif's prevailing wind comes from the north-west. An air tower filling the leftover tip of the north-east cut corner (1.0 m deep, about 0.9 m^2 clear: enough for one person to step in and clean, with the main cleaning from a roof hatch), so it takes no usable room space, catches it above the roof, where it is strong and clean, and brings it down to a tunnel below the floor. The tunnel passes through the cool ground and releases the air through a low, closable opening in the courtyard wall. Warm air rises and leaves through the open sky. The narrow courtyard is enclosed on all four sides and shaded by its walls, so it receives air without receiving sun. Every opening can be closed: open at night and early morning, closed in hot afternoons, dust and cold.

On firm ground, as far north as possible

The house sits on the firm, higher ground in the north of the plot, square to the M line so every wall can be set out from the site pegs, and pushed as far north as the setbacks allow. The narrow courtyard keeps the house narrow, and the north-east corner is cut parallel to the lane, so the whole house reaches further north and both palms stay well clear behind it.

One living room, two heights

The living room is one open space with two ceiling heights. The half next to the courtyard is double height: it continues the courtyard's vertical space, gives the best view into it, and works as a chimney, with high operable windows that let warm air rise and leave while cool air is drawn in from the courtyard; its 4.70 m axis width (4.50 m clear) is a simple roof span. The east half is normal height: a warmer, lower place for dining beside the kitchen door, easier to air-condition and light, with ducts hidden in its ceiling, and the first floor continues above it. No upper floor is built over the double-height half or the courtyard.

One band that serves the house

Everything that serves the house sits in one band on the west: the family entrance with its entry cupboard, the future lift shaft, the stair, the utility room with the service shaft and the west wash block, one strip of wet rooms on one drain run (hospitality kitchen, family basins, family WC, guest WC, guest basins). Pipes, cables and air rise in one place, the structure gains one stiff core, and every service can be reached for maintenance without breaking a wall. The family lobby and riwaq run along this band as one straight strip on the courtyard side, the family's covered route from the front door to the living room.

Family, guests and service, each in their place

The family enters through the lobby; women guests use the same door to reach the women's majlis. The men's majlis has its own foyer, entered by a double door at one end and opening into the majlis at the other (a bent entry), with a shoe cabinet and hooks for the bisht and shemagh, a trough basin with three taps and a WC whose door hides behind a short screen wall, all in the south of the west wash block; the guest side has no door onto the riwaq. Its only link to the house is a short serving passage with only two doors that are never open together: the family opens door 1 from the riwaq, leaves the tray and closes it, then calls the host, who opens door 2 from the guest passage and collects. Door 1 locks from the riwaq side and door 2 from inside the passage, so the family controls both. An open door 2 shows only the passage and a blank wall, a bent entry (majaz). The hospitality kitchen just north of it, a light kitchen for coffee, tea, dates and desserts, is the serving base; the east kitchen is the heavy one. The living room opens to the riwaq without a door and only a 1.20 m low partition faces the serving door, so the family's privacy rests on the two serving doors never being open together. The kitchen stays isolated beside the courtyard, a clean 4.36 × 6.5 m room on the same lines as the courtyard, with two parallel counter runs, its only door to the dining and an open entrance to the laundry. The laundry runs north along the solid wall of the women's majlis to the maid room and maid bath, which use the sloped corner to the full; the maid wing is reached only through the kitchen. Kitchen, laundry and maid bath drain straight out to the east, to a manhole in the east service yard; the vent stack runs up the outside face of the east wall, and the kitchen extract leaves straight through the east wall at the hood (V5.0: the east duct is removed).

Services built for life

The service shaft sits inside a utility room, so every service is in one place behind one door: the shaft, the main electrical panel, the water manifold with its isolation valves and the water heater. The shaft is a closed box of about 1.0 × 1.2 m clear with sealed walls and one full-height maintenance door, and the 0.9 m in front of that door is always kept clear. It stands directly beside the west wash block, so drainage runs stay short; first-floor bathrooms should stack over the north part of the block, near the shaft. The block's drain crosses under the serving passage floor, with a cleanout access point there. The space under the stair's landing becomes a store instead of dead space. The lift shaft is built now as a store and fitted with a two-stop lift later. The Saudi Building Code sets no fixed size for a closed pipe shaft; the sizes here follow general professional guidance and must be confirmed with the code, the municipality and the plumbing engineer.

Concept for discussion. Airflow, tunnel depth, waterproofing, drainage, structure and services need the engineers' review; wind direction to be confirmed on site.